Google Skills

When you need memory bank

Good use cases

- Customer service agents (remember past issues)
- Personal assistants (remember preferences)
- Learning systems (remember what user has learned)
- Personalization (remember interests)

When session state is enough

- Simple Q&A agents (no memory needed)
- Task-focused agents (complete and forget)
- Short interactions (single-purpose)

Resources for learning more

When you're ready to implement Memory Bank:

Official documentation

- **Memory bank overview:** https://cloud.google.com/vertex-ai/generative-ai/docs/agent-engine/memory-bank/overview
- **ADK documentation:** https://google.github.io/adk-docs/

Implementation details

The ADK documentation covers:

- VertexAiMemoryBankService configuration
- Memory ingestion callbacks
- PreloadMemoryTool usage
- Express Mode for testing

Key takeaways

You now understand how memory bank complements session state to create a complete memory architecture. While session state handles the current conversation, memory bank enables agents to learn and remember across all interactions—a key capability for personalization and long-term user engagement.

Session state vs memory bank

- **Session State:** Current conversation (course 4)
- **Memory Bank:** Cross-session learning (course 9)
- **Together:** Complete memory architecture

Memory bank basics

- LLM-powered extraction (intelligent, not raw)
- Semantic search (understands intent)
- Managed service (no setup needed)

When to use

Need	Solution
Conversation context	Session state
Cross-session learning	Memory bank
Both	Use both together